

Universal Wells Application - Setup Guide

Overview

This refactored system allows you to manage multiple monitoring wells from a single codebase, making it easy to add new wells and maintain consistent functionality across all locations.

Directory Structure

```
/your-root-directory/
├── wells_config.php    # Central configuration for all wells
├── index.php           # Root index showing all available wells
├── css/
│   └── styles.css      # Shared styles
├── hp/                # Horse Park well
│   ├── config.php      # Loads well-specific config
│   ├── api.php         # API functions
│   ├── index.php       # Main well page
│   ├── monthly.php     # Monthly data view
│   └── fetch_data.php  # AJAX endpoint
├── hickman1/          # Hickman1 well
│   ├── config.php      # Same file, auto-detects well
│   ├── api.php         # Same API functions
│   ├── index.php       # Same structure
│   ├── monthly.php     # Same structure
│   └── fetch_data.php  # Same structure
└── [other-wells]/     # Additional wells follow same pattern
```

Step-by-Step Setup

1. Create the Central Configuration

Create `wells_config.php` in your root directory using the provided code. This file contains all well configurations in one place.

2. Add Your Wells to Configuration

Edit `wells_config.php` and add entries for each well:

```
php

'hickman1' => [
    'well_id' => 'hickman1',
    'location_id' => '1234567890123456', // Get from HydroVu API
    'common_name' => 'Hickman1',
    'full_name' => 'Hickman 1 Water Well',
    'water_level_baseline' => 750.00, // Well-specific baseline
    'water_level_warning' => 780.00, // Ground surface elevation
    'description' => 'Hickman 1 groundwater monitoring well',
    'param_names' => [
        'depth' => 'Groundwater Level Elevation',
        'temperature' => 'Temperature'
    ],
    'param_units' => [
        'depth' => 'ft',
        'temperature' => '°F'
    ]
]
```

3. Create Directory for Each Well

For each well, create a directory named after the `well_id`:

- `/hp/` for Horse Park (already exists)
- `/hickman1/` for Hickman1 (new)
- etc.

4. Copy Files to Each Well Directory

Copy these files from `/hp/` to your new well directory:

- `config.php` (updated version from artifacts)
- `api.php` (existing file, no changes needed)
- `index.php` (updated version with loading skeleton)
- `monthly.php` (update similar to index.php)
- `fetch_data.php` (existing file, no changes needed)

Important: The updated `config.php` automatically detects which well